

shear strength. They design construction plans for the proposed building taking into account these parameters.

Latest trends in geological T&M

The amount of data collected is increasing at a spectacular rate.

Satish Mohanram, technical marketing manager, National Instruments-India, elaborates, "Scientists and engineers doing test and measurement end up collecting huge amounts of data. To give some examples, seismic information taken at high sampling rates, structural studies done on historical monuments, property testing of objects, etc create data, and making sense out of this information is the objective of the exercise. This could become intimidating, considering the amount of data collected."

Another emerging trend in geotechnical investigation is a greater emphasis on *in-situ* tests. *In-situ* literally means 'in position.' *In-situ* means tests conducted on the site. These tests can be effectively used to predict foundation behaviour with a high reliability.

Products developed for regulated industries such as automotive, aerospace and medical must comply with rigorous development standards and certifications. Revisions to these standards place increased scrutiny on the quality and accuracy of test tools—creating an increased burden of proof to demonstrate that testers have been qualified for use. Though businesses outside these industries will not feel an immediate impact from these trends, they can benefit from detecting defects earlier in the life cycle and drive down product development cost.

Geotechnical investigation is gaining prominence for a number of reasons. Main among them is the increased activity in construction—not just in terms of volume but also complexity. Gone are the days when soil was a decisive factor determining the kind of construction to be done at a place. Nowadays, the soil is made suitable for the project

Tools and technology used in geological testing

- Dropping pipettes like digital micropipettes are used for testing as these provide a number of additional features over traditional pipettes, including larger holding area for fatigue-free operation, and digital counters as well as push-button sample extraction. The Nichipet EX autoclavable digital micro pipette is one such tool that also ensures that testing is not affected by the temperature of the technician's hand.
- Magnetometer geophysical instruments like cryogenic magnetometers, proton magnetometers and spinner magnetometers are used to detect ferrous metals at a very large depth. Newer quantum magnetometers take advantage of the spin of subatomic particles (nuclei and unpaired valence electrons). Through a process of polarisation, particles are caused to precess in the earth's ambient magnetic field. The resulting frequency of precession can be translated directly into magnetic field units. Quantum results are scalar (total field intensity) as opposed to vector (i.e., from fluxgate geophysical instruments or GEM's suspended dIdD technology).
- Rugged water-level indicators are now available, protected in Kevlar and featuring sturdy cables incorporating multiple conductors to fit easily into standpipes and wells. Firms like Durham Geo also supply laser-marked cables for high-contrast measurements that do not wear off with use.
- Downhole cameras are used to inspect problems encountered in well drilling. These cameras are heavy-duty oceanographic-type cameras used for high pressure deployment. The camera is suspended on an aluminium heavy-duty cable reel so that data can be carried to the technician without sacrificing the strength of the cable. Naeva Geophysics' Well-Vu WV-C1000 comes with a tri-legged stand and offers real-time video output to the field computer.
- Analytical or scientific software like Landmark Graphics GeoGraphix software, Parallel Geoscience SPW software and Seismic Micro-Technology KINGDOM are also available.
- Computer-aided design (CAD) software like Autodesk AutoCAD and Midland Valley 2DMove are also used by geological test technicians.

Use of test automation software has increased rapidly over the last decade due to the need for highly customisable, flexible and capable measurement systems. Software-centric test solutions are the only viable approach for delivering complex technologies under aggressive timelines, limited resources and constant product churn.

Mohanram says, "Test and measurement, in today's scenario, is moving more towards a software-defined approach. Geological T&M is no exception to it as it enables scientists to make the best use of the time and effort that they put in taking these measurements. This approach enables engineers to get

more flexibility and greater performance at lower cost. The innovative platform-based Graphical System Design approach enables this new paradigm of test and measurement."

Increasing product complexity and capability has a direct impact on the reliability, performance and accuracy of test systems. Consequently, "there is increased focus on ensuring quality and reliability of test software through life-cycle management and development practices formerly reserved for embedded systems. Some organisations are voluntarily applying these development practices to improve test software and build more feature-rich and defect-free test solutions, but a growing number of industries will be required to use similar practices to comply with regulatory standards," explains a senior engineer from this field.

These standards set a high bar for process and quality, but best practices in software engineering ensure that test systems meet increasingly demanding feature and performance requirements.